

Assignment 1: Tetris Puzzle

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1. What interesting **properties** does your polyomino piece have within the systems of Tetris? Why?

My new Tetris piece “The U piece” has a larger body than the other base tetris pieces. This results from having more cells that make up its body. The U piece also has a skirt that differs based on the rotation of the piece. Three of out the four rotations have a skirt made up of three cells, while the fourth rotation has a skirt made up of only two cells with a gap in the middle. All of these rotations are distinct from one another, unlike the I or O pieces. These distinct rotations create lots of new and different opportunities for the player that the current tetris pieces do not already allow for. The U piece kinda can be looked at as an opposite to the O piece. While the O piece is 2x2 and solid, the U piece is 3x3 with a 2 cell opening from one side. This creates different opportunities that no other piece can really provide.

2. What interesting **relationships** does your piece sequence have with the board state within the envisioned play session? Why?

Some interesting relationships that my piece sequence has within the created board state is using the two “prongs” that are the top of the U, you can fill two holes that are one cell apart and are two deep at the same time. This can result in either setting up being able to clear multiple lines or can clear multiple lines by itself. When turned on its side either way, the two prongs again can allow for multiple rows to be cleared at the same time. But when doing so it will leave behind the bottom middle cell. This can allow for strategic gameplay opportunities, where the player can think about which side to leave the cell on based on which piece that they have coming up next. The two cell space in the middle of the U also has a good relationship with the L, J, and I pieces because they can all fit the two cell long parts of them into the two cell deep part of the U piece.

3. How do you expect your board state to provide an interesting challenge to players? **Consider objects, properties, behaviours, and relationships.**

I expect my board state to provide an interesting challenge to players because of the uniqueness of what the U piece can do and the value that it can provide. With the ability

to clear multiple lines and in specific orientations leave pieces behind that can be used in conjunction with other pieces, thus giving the player the ability to clear another line or create an opportunity to set up their board better to be able to place the right piece to maximize for the most points. The ability to clear multiple lines at once is also very useful in a Big Bang Tetris puzzle because it can clear lines very quickly but also with the uniqueness of this piece you can place it and clear a single line while setting yourself up to clear a line later in the game.

